

MANASA RAPURU

Bioinformatics Engineer | <https://manasarapurur.github.io/portfolio/>

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SUMMARY

- Bioinformatics engineer experienced in building scientific applications, analytical pipelines and AI-powered tools across genomics, transcriptomics, and molecular diagnostics
- Experienced in working closely with R&D, product, and engineering teams to make complex data accessible and actionable.

EDUCATION & CERTIFICATIONS

M.S Bioinformatics @ Boston University	Sept 2021 - May 2023
B.S. Microbiology & Chemistry Minor @ San Jose State University	Aug 2016 - Dec 2020
AWS Cloud Practitioner Certification	In Progress
AI Developer Certification @ Coursera	Jan 2026
Full Stack Developer Certification @ Coursera	April 2024

EXPERTISE

- **Bioinformatics**
 - **Technologies:** RNA-seq, scRNA-seq, CRISPR, qPCR/dPCR Assay Design, WES, WGS, Phylogenetics, Protein Modeling
 - **Tools:** Samtools, Bedtools, STAR, IGV, UCSC Genome Browser, Ensembl, NCBI, PyMOL, Clustal Omega, MUSCLE, Mugsy, Bowtie2, Tophat, COSMIC, dbSNP, BLAST, GATK, FastQC, Trimmomatic
- **Scientific Software Development**
 - **AI & Machine Learning:** NLP, RAG, Knowledge Graphs, Prompt Engineering, Claude Code, GitHub Copilot, Embeddings, Vector Databases, LLM APIs, AI agents
 - **Programming & Data Science:** Python, R, SQL, Bash, Java, C++, Perl, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Git
 - **Backend:** FastAPI, Swagger Flask, Django, REST API, JSON, Streamlit
 - **Cloud & Infrastructure:** AWS, Docker, HPC, Linux (Red Hat/CentOS), SLURM, CI/CD
 - **Front End:** React, HTML, CSS, JavaScript, TypeScript, Tailwind CSS, Material UI, Streamlit

PAST WORK

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| Bioinformatics Scientist at ThermoFisher Scientific
(qPCR/dPCR TaqMan Assays) | Apr 2024 - Nov 2025 |
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- **Scientific Software Development & AI**
 - Defined product requirements with cross-functional stakeholders and translated scientific workflows into application interfaces to improve operational efficiency, reduce execution time and resource overhead. Also developed and deployed service to be accessible across multiple internal teams.

- Architected and developed a conversational AI chatbot leveraging NLP and retrieval-augmented generation (RAG) during exploration of low-efforts that could increase accessibility to internal application-based data.
- **Bioinformatics**
 - Contributed to a revenue-generating commercial service by delivering custom TaqMan Assay assay designs for customers' molecular diagnostics and targeted genomic analysis initiatives.

Bioinformatics Specialist at Massachusetts General Hospital **Mar 2021 - Aug 2023**
(Huntington's Disease)

- **Scientific Software Development & Machine Learning**
 - Developed a Python-based analytical pipeline (Pandas, NumPy, Matplotlib, Scikit-Learn) to model cell morphology metrics derived from IncuCyte imaging.
 - Evaluated machine learning approaches to assess whether custom models could improve phenotypic classification accuracy in comparison to the instrument's default analysis software.
- **Bioinformatics**
 - Conducted transcriptomic analysis of RNA-Seq, Single-cell RNA-Seq, and tandem repeat genomic datasets to investigate disease-associated gene expression patterns and molecular signatures in Huntington's disease, implemented differential expression workflows in R using DESeq2 and ggplot2.
 - Engineered automated genomic data processing pipelines using Bash for execution on HPC clusters, configuring computational environments on CentOS and integrating precompiled and custom-built C/C++ bioinformatics tools.

Bioinformatics Master's Student at Boston University **Sept 2021 - May 2023**

- **Scientific Software Development**
 - Designed and developed a full-stack web application to support exploration of postmortem brain transcriptomic data from Alzheimer's disease research. Provided dynamic visualizations of differential gene expression, CPM values, pathway activity across brain regions and disease types. Implemented the platform using HTML, CSS, JavaScript, Python, and SQL.
- **Bioinformatics**
 - Analyzed RNA-seq, microarray, ChIP-seq, and single-cell datasets using R and automated Snakemake workflows.

Bioinformatics Intern at Sandia National Laboratories **Jun 2020 - Aug 2020**
(Phage Therapy & Phylogenetics)

- **Bioinformatics**
 - Conducted computational genomics analyses supporting Phage Therapy research, identified integrative genetic elements in bacterial genomes through phylogenetic and comparative analyses.

- Executed and optimized bioinformatics workflows on Red Hat Linux–based HPC clusters, managing job scheduling and troubleshooting memory and system I/O constraints.
- Debugged and enhanced Perl and Python bioinformatics pipelines, contributing improvements to open-source repositories using Git.

Undergraduate Bioinformatics Research Student
(Candidatus Saccharibacteria)

Jan 2019 - Dec 2020

- **Bioinformatics**

- Conducted phylogenetic and protein visualization analyses to understand evolutionary relationships and potential functional role of Candidatus Saccharibacteria in the oral microbiome
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